Chain Supply Analysis

Power BI Dashboard Assignment
Role: Data Analyst
Objective: Analyze orders, sales, profitability, customers, and shipment performance using the supplied chain supply dataset.

1. Business Objective

Build an interactive Power BI dashboard that helps management understand overall sales performance, profitability, customer markets, product performance, and supply chain delivery efficiency. The dashboard should convert the available order and shipment information into clear business insights.

2. Key Metrics to Calculate

Metric	Purpose
Total Orders	Count of unique Order ID values.
Total Order Items	Count of Order Item ID records.
Total Quantity	Sum of Order Quantity.
Total Gross Sales	Sum of Gross Sales.
Total Profit	Sum of Profit.
Average Order Value	Total Gross Sales divided by unique orders.
Average Order Quantity	Total Order Quantity divided by unique orders.
Average Profit per Order	Total Profit divided by unique orders.
Average Discount %	Average of Discount %.
Average Shipment Days	Average of Order and Shipment diff.
Scheduled Shipment Days	Average or total scheduled shipment days, as appropriate.
On-Time Shipment Rate	Percentage of records where actual shipment days are within scheduled shipment days.
Late Shipment Rate	Percentage of records where actual shipment days exceed scheduled shipment days.
Profit Margin %	Total Profit divided by Total Gross Sales × 100.

3. Recommended Dashboard Sections

Section	Analysis Required
Executive Summary	Total Orders, Total Quantity, Gross Sales, Profit, Profit Margin, Average Order Value, Average Shipment Days.
Sales Analysis	Sales and profit by Order Year, Order Month, Customer Market, Customer Region, and Product Department.
Product Analysis	Sales, quantity, profit, discount, and profit margin by Product Department, Product Category, and Product Name.
Customer Analysis	Unique customers, orders, sales, profit, and average order value by Customer Market, Region, and Country.
Shipment Analysis	Shipment performance by Shipment Mode, Warehouse Country, Customer Market, and time period.
Geographical Analysis	Sales, profit, orders, and shipment performance by Customer Country and Warehouse Country.
Time Analysis	Monthly and yearly trends for orders, sales, profit, quantity, and shipment days.

4. Required Visuals

Visual	Suggested Fields / Purpose
KPI Cards	Total Orders, Total Sales, Total Profit, Profit Margin %, Total Quantity, Average Shipment Days.
Line Chart	Monthly Sales and Profit trend.
Column Chart	Sales and Profit by Product Category.
Bar Chart	Top Product Names by Sales or Profit.
Stacked Column / Bar	Sales by Customer Market and Product Department.
Map	Sales or Profit by Customer Country.
Donut / Bar	Order or sales distribution by Shipment Mode.
Column Chart	Shipment performance by Warehouse Country.
Matrix	Customer Region → Customer Country with Orders, Sales, Profit, Margin, and Shipment Days.
Scatter Chart	Gross Sales vs Profit by Product or Category; use Quantity or Discount as a supporting field.
Slicers	Order Year, Order Month, Customer Market, Customer Region, Product Department, Product Category, Shipment Mod

5. Important Business Questions

• Which years and months generated the highest sales and profit?

• Which product departments and categories contribute the most revenue and profit?

• Which products have high sales but comparatively low profit?

• Which customer markets, regions, and countries generate the highest sales?

• Which markets generate the highest profit margin?

• How does discount percentage relate to profit?

• Which shipment modes are used most frequently?

• Which shipment modes or warehouse countries have higher delivery delays?

• Which countries or regions have the highest average shipment time?

• Are shipment delays increasing or decreasing over time?

• Which customers have high order value or high order frequency?

• Are there markets where sales are high but profitability is low?

6. Data Preparation Requirements

Before creating visuals, inspect the data types and data quality. Convert Order Date and Shipment Date to Date type, Order Time to Time type, and numeric fields such as Order Quantity, Gross Sales, Discount %, Profit, and shipment-day fields to appropriate numeric types. Check for blanks, duplicate Order Item ID values, invalid dates, inconsistent category names, and unexpected negative or zero values.

Create calculated measures rather than relying only on raw columns for the main KPIs. Where a metric uses Order ID, consider whether a distinct count is required because the same order can contain multiple order items.

7. Suggested Power BI Measures

Measure	Business Calculation
Total Orders	DISTINCTCOUNT(Order ID)
Total Order Items	COUNT(Order Item ID)
Total Quantity	SUM(Order Quantity)
Total Gross Sales	SUM(Gross Sales)

Measure	Business Calculation
Total Profit	SUM(Profit)
Average Order Value	Total Gross Sales / Total Orders
Profit Margin %	Total Profit / Total Gross Sales
Average Shipment Days	AVERAGE(Order and Shipment diff)
Late Shipment Rate	Late shipments / total shipment records

8. Dashboard Expectations

• Create a clean, professional dashboard with a clear title and logical arrangement of visuals.

• Use consistent number formatting: currency for sales and profit, percentage for discount and margin, and whole numbers for quantity and orders.

• Provide meaningful slicers so the user can analyze different years, markets, regions, categories, and shipment modes.

• Use meaningful chart titles that explain what is being measured.

• Avoid overcrowding the page. Group related KPIs and visuals together.

• Use drill-down where useful, especially for Date, Region/Country, and Product hierarchy.

• Add tooltips where they improve interpretation.

• Validate totals in the dashboard against the source data before submission.

9. Final Deliverable

Submit the Power BI (.pbix) file containing the completed Chain Supply Analysis dashboard. The final report should allow a business user to move from an overall performance view to detailed analysis of products, customers, geography, time trends, and shipment performance.

Submission focus: Correct calculations, relevant KPIs, appropriate visual selection, clear layout, interactive filtering, and useful business observations derived from the data.